Algebra Monsoon 2026 Lecture 4

Free magmas

Example 1 Consider pairs (*) (M, x) consisting of a monoid M and an element $x \in M$. A morphism of such pairs (**)

$$\varphi : (M, x) \longrightarrow (N, y)$$

is a morphism of monoids $\varphi : M \to N$ such that $\varphi(x) = y$. We ask:

Is there a pair (L, α) having the universal property that for every pair (M, x), there is a unique morphism $(L, \alpha) \longrightarrow (M, x)$?

Yes indeed; $(\mathbb{N}, 1)$ is one such pair, as we have seen.

Note: • $Id_M : (M, x) \to (M, x)$ is a morphism

• If f, g are morphisms, then so is $g \circ f$.

as we have ... got.

Exercise: Show that the only morphism of monoids $\varphi : \mathbb{N} \to \mathbb{N}$ such that $\varphi(1) = 1$ is $\varphi = Id_{\mathbb{N}}$.

(*) Maybe such pair should be called a pointed monoid.

(**) morphism of pointed monoids

Example 2 Consider pairs (*) (G, x) consisting of a group G and an element $x \in G$. A morphism* of pairs (**)

$$\varphi : (G, x) \to (H, y)$$

is a morphism of groups $\varphi : G \to H$ such that $\varphi(x) = y$. We ask:

Is there a pair (L, α) having the universal property that for every pair (G, x), there is a unique morphism of pairs $(L, \alpha) \to (G, x)$?

Yes indeed; we have seen that $(\mathbb{Z}, 1)$ is one such pair.

Actually, $(\mathbb{Z}, -1)$ is another such pair.

⊛ pointed group

⊛⊛ or morphism of pointed groups

⊛ • $Id_G : (G, x) \to (G, x)$ is a morphism

• If $(G_1, x_1) \xrightarrow{f_1} (G_2, x_2) \xrightarrow{f_2} (G_3, x_3)$ are morphisms then so is $f_2 \circ f_1$.

Exercise By the universal property of $(\mathbb{Z}, 1)$, there is a unique morphism of groups $\varphi : \mathbb{Z} \to \mathbb{Z}$ such that $\varphi(1) = -1$.
Show that φ is an isomorphism of groups.

Let S be a set. Consider pairs

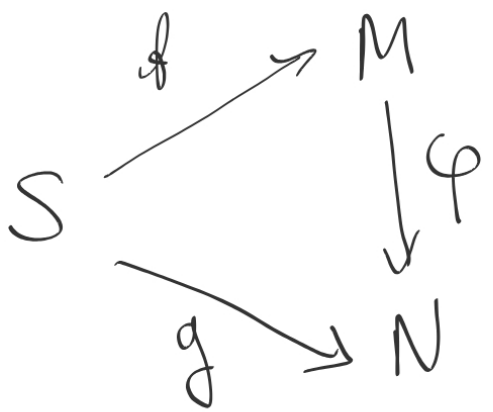
$$(M, f)$$

where M is a magma and $f : S \to M$ is a mapping.

By a morphism of pairs

$$\varphi : (M, f) \longrightarrow (N, g)$$

we mean a morphism of magmas $\varphi : M \to N$ such that the triangle



Commutes (ie. $g = \varphi \circ f$).

It is clear that

- $Id_M : (M, f) \longrightarrow (M, f)$ is a morphism of pairs.
- If $\varphi_1 : (M_1, f_1) \longrightarrow (M_2, f_2)$ and $\varphi_2 : (M_2, f_2) \longrightarrow (M_3, f_3)$ are morphisms of pairs, then so is

$$\varphi_2 \circ \varphi_1 : (M_1, f_1) \longrightarrow (M_3, f_3).$$

We say that a morphism of pairs

$$\varphi : (M, f) \longrightarrow (N, g)$$

is an <u>isomorphism</u> if there exists a morphism of pairs

$$\psi : (N, g) \longrightarrow (M, f)$$

such that

$$\psi \circ \varphi = Id_M$$

and $\varphi \circ \psi = Id_N$.

We ask: Is there a pair (L, i) having the universal property that for every pair (M, f), there is a unique morphism of pairs

$$(L, i) \longrightarrow (M, f) \; ?$$

It is not obvious if such an initial pair (also called the free magma on the set S) exists.

Proposition If (L, i) is an initial pair, then the only morphism $(L, i) \longrightarrow (L, i)$ is Id_L.

Proof Id_L is certainly such a morphism. If $\varphi : (L, i) \longrightarrow (L, i)$ is any morphism, then $\varphi = Id_L$ by the universal property. $\square$

Proposition If (L_1, i_1) and (L_2, i_2) are two initial pairs, then the unique morphism

$$\varphi : (L_1, i_1) \longrightarrow (L_2, i_2)$$

is an isomorphism of pairs.

Proof Since (L_2, i_2) is an initial pair, there is a unique morphism of pairs

$$\psi : (L_2, i_2) \longrightarrow (L_1, i_1).$$

Now $\psi \circ \varphi$ and $\varphi \circ \psi$ are morphisms of initial pairs, so (by a previous prop)

$$\psi \circ \varphi = Id_{L_1} \quad \text{and} \quad \varphi \circ \psi = Id_{L_2}.$$

It follows that φ is an isomorphism of pairs. □

Summary: If an initial pair exists, it is unique up to a unique isomorphism.

Construction of an initial pair :

Given a set S, define

$$L_1(S) = S, \quad L_2(S) = S \times S$$

and inductively

$$L_{n+1}(S) = \coprod_{1 \le i,j \le n} S_i \times S_j \qquad (n \ge 1).$$

Put $L(S) = \coprod_{n \ge 1} L_n(S)$.

Explicitly, the elements of $L_1(S)$ are

(s), where $s \in S$;

the elements of $L_2(S)$ are

all pairs (s,t), where $s, t \in S$.

The elements of $L_3(S)$ are (of two types)

$((s,t),u)$, $(s,(t,u))$

where $s,t,u \in S$. The elements of

$L_4(S)$ are (of five types)

$$((s,t),u),v), \quad ((s,t),(u,v)), \quad (s,((t,u),v))$$

$$((s,(t,u)),v), \quad (s,(t,(u,v)))$$

where $s,t,u,v \in S$, and so on.

The definition $L(S) = \coprod_{n \geq 1} L_n(S)$ means that for every $x \in L(S)$, there is a unique $n \geq 1$ such that $x \in L_n(S)$. (we view each $L_n(S)$ as a subset of $L(S)$).

The mapping $i : S \longrightarrow L(S)$ is just the inclusion of $S = L_1(S)$ in $L(S)$.

We have used the notation $i(s) = (s)$ for every $s \in S$.

The law of composition in $L(S)$.

Let $x, y \in L(S)$. There are unique integers $m, n \in \mathbb{N}_{\geq 1}$ such that

$$x \in L_m(S) \quad \text{and} \quad y \in L_n(S).$$

We pose $xy = (x, y)$ in $L_{m+n}(S)$.

Example If $x = (a, b)$ for $a, b \in S$ and $y = ((c, d), e)$ for $c, d, e \in S$, then $m = 2$, $n = 3$, and

$$xy = \big((a, b), ((c, d), e)\big) \quad \text{in } L_5(S).$$

This law of composition makes $L(S)$ into a magma and $(L(S), i)$ into the kind of pair we are considering.
Note that the magma $L(S)$ is generated by $i_*(S)$.

Proposition The magma $L(S)$ is generated by the subset $i(S) \subset L(S)$.

Proof: This is clear from the construction: use induction to show that $L_n(S) \subset \langle i(S) \rangle$, starting from $L_1(S) \subset \langle i(S) \rangle$, which is given. □

Proposition $(L(S), i)$ is a free magma on the set S.

Proof (Sketch) We have to show that $(L(S), i)$ has the universal property that for every pair (M, f) consisting of a magma M and a mapping $f : S \to M$, there is a unique morphism of pairs $\varphi : (L(S), i) \to (M, f)$. Equivalently, there is a unique morphism of monoids $\varphi : L(S) \to M$ such that

$$f = \varphi \circ i .$$

We define φ on $L_n(S)$ by induction on n. If $n = 1$, every element of $L_1(S)$ is of the form $i(s)$ for a unique $s \in S$, and we put $\varphi(i(s)) = f(s)$.

Suppose that φ has been defined on $L_r(S)$ for every $r \in [1, n]$ for some $n \geq 1$, and let us define it on $L_{n+1}(S)$. Every element $x \in L_{n+1}(S)$ is uniquely of the form

$$x = (y, z),$$

where $y \in L_i(S)$ and $z \in L_j(S)$ for some $i, j \in [1, n]$ such that $i + j = n + 1$. Define

$$\varphi(x) = \varphi(y)\varphi(z).$$

It is clear by construction that $\varphi : L(S) \to M$ is a morphism of magmas and $f = \varphi \circ \iota$.

As to the uniqueness of φ, let $\psi : L(S) \to M$ be another morphism of magmas such that $f = \psi \circ i$. This last requirement means that φ and ψ agree on $L_1(S)$. Let us prove that φ and ψ agree on $L_n(S)$ by induction on n; the case $n = 1$ has already been done. Suppose that φ and ψ agree on $L_r(S)$ for every $r \in [1, n]$ and some $n \geq 1$. Every $x \in L_{n+1}(S)$ is uniquely of the form

$$x = (y, z) = yz,$$

where $y \in L_i(S)$ and $z \in L_j(S)$ for some $i, j \in [1, n]$ such that $i + j = n + 1$.

We have

$$\begin{aligned}\varphi(x) &= \varphi(yz) \\ &= \varphi(y)\varphi(z) \\ &= \psi(y)\psi(z) \\ &= \psi(yz) \\ &= \psi(x) \quad ,\end{aligned}$$

because φ and ψ agree on $L_i(S)$ and $L_j(S)$. By induction, $\psi = \varphi$, because $L(S)$ is the union of the $L_n(S)$, $n \geq 1$ $\square$

The uniqueness also follows from the fact proved earlier that if two morphisms of magmas agree on a generating set, then they are equal. Alternately, one can derive that result from the proof given here.

Definition The initial pair $(L(S), \iota)$ (or just the magma $L(S)$) is called a (or the) free magma on the set S.

Notation $F_{\mathcal{L}}(S)$ = the free magma on S

- $F_{\mathcal{L}}(\emptyset) = \emptyset$
- $F_{\mathcal{L}}(\{1\}) = ?$
- $F_{\mathcal{L}}(\{a, b\}) = ?$

Proposition Let M be a magma, $S \subset M$ a subset, and $\varphi : F_{\mathcal{L}}(S) \to M$ the unique morphism of magmas such that $\varphi(s) = s$ for every $s \in S$. We have

$$\langle S \rangle = M \iff \varphi \text{ is surjective.}$$

For every pair (M, f), denote by

$$\varphi_f : F_{\mathcal{L}}(S) \longrightarrow M$$

the unique morphism of magmas such that $f = \varphi_f \circ i$.

Proposition Let S be a set, M a magma. The mapping

$$\mathrm{Hom}_{\mathcal{E}}(S, M) \longrightarrow \mathrm{Hom}_{\mathcal{L}}(F_{\mathcal{L}}(S), M)$$

($\mathcal{E}$: sets; $\mathcal{L}$: magmas)

sending a mapping $f \in \mathrm{Hom}_{\mathcal{E}}(S, M)$ to the unique morphism of magmas

$$\varphi_f : F_{\mathcal{L}}(S) \longrightarrow M$$

such that $f = \varphi_f \circ i$ is a bijection.

such.

Proof The mapping $f \longmapsto \varphi_f$ is injective because f can be recovered from φ_f as $f = \varphi_f \circ \iota$.

It is surjective because, for every $\varphi \in \mathrm{Hom}_{\mathcal{L}}(L(S), M)$, we have

$$\varphi = \varphi_f, \text{ where } f \in \mathrm{Hom}_{\mathcal{E}}(S, M)$$

is the mapping $f = \varphi \circ \iota$. □

Def A magma is called a free magma if it is isomorphic to $F_{\mathcal{L}}(S)$ for some set S.

Exercise: Show that every magma is a quotient of a free magma.

Exercise: Think of a way of attaching a morphism of magmas

$$F_{\mathcal{L}}(f) : F_{\mathcal{L}}(S) \longrightarrow F_{\mathcal{L}}(T)$$

to a mapping of sets $f : S \to T$.

What is $F_{\mathcal{L}}(Id_S)$?

If $g : T \to U$ is another mapping, what is the relationship between $F_{\mathcal{L}}(f)$, $F_{\mathcal{L}}(g)$, and $F_{\mathcal{L}}(g \circ f)$?

Let $\beta : M \to N$ be

Proposition Let $T \xrightarrow{\alpha} S$ and $M \xrightarrow{\beta} N$ be a mapping and a morphism of magmas.

The diagram

$$\begin{array}{ccc}
\mathrm{Hom}_{\mathcal{E}}(S, M) & \xrightarrow{\sim} & \mathrm{Hom}_{m}(F_{\mathcal{L}}(S), M) \\
{\scriptstyle \circ\alpha}\downarrow & & \downarrow{\scriptstyle \circ F_{\mathcal{L}}(\alpha)} \\
\mathrm{Hom}_{\mathcal{E}}(T, M) & \xrightarrow{\sim} & \mathrm{Hom}_{m}(F_{\mathcal{L}}(T), M) \\
{\scriptstyle \beta\circ}\downarrow & & \downarrow{\scriptstyle \beta\circ} \\
\mathrm{Hom}_{\mathcal{E}}(T, N) & \longrightarrow & \mathrm{Hom}_{m}(F_{\mathcal{L}}(T), N)
\end{array}$$

is commutative.

Exercise 1) How would you define the free unital magma $(F_u(S), j)$ on a set S?

2) How would you construct it?

2) Take $F_u(S) = \{e\} \amalg F_a(S)$.
(where $\{e\} = S^0$).

Exercise How would you define the free associative magma $(F_g(S), i)$ on a set S?

Take $F_g(S)$ to be the set of all sequences

$$(s_1, \dots, s_m) \qquad (m \geq 1)$$

of elements of S and define

$$(s_1, \dots, s_m)(t_1, \dots, t_n)$$

$$= (s_1, \dots, s_m, t_1, \dots, t_n).$$

The free monoid on S is

$$F_m = \{e\} \amalg F_g(S)$$

where $\{e\} = S^0$.

See the next lecture.

Exercise Let S be a set and $F_{\mathcal{L}}(S)$ the free magma on S.

Let C be the smallest congruence on $F_{\mathcal{L}}(S)$ such that

$$(xy)z \sim_C x(yz)$$

for all $x, y, z \in (S)$. Show that $F_{\mathcal{L}}(S)/C$ is a free associative magma on S. Make the resulting isomorphism

$$F_{\mathcal{L}}(S)/C \xrightarrow{\iota} F_{\mathcal{G}}(S)$$

explicit (where $F_{\mathcal{G}}(S)$ is the free associative magma consisting of all sequences of elements of S of length ≥ 1).